Observability: Can I estimate an initial state?

- We define a system to be observable if we can determine its initial state vector $x(0)$ via processing its input signal $u(t)$ and its output signal $z(t)$.
- Since we can simulate the system's state and output if we know $x(0)$ and $u(t)$ this also implies that we can determine $x(t)$ for all $t \geq 0$:

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)\,\mathrm{d}\tau.$$

- So, it should not be surprising that a system must be observable for the KF to work (noting an exception to this rule that is stated on the summary slide).
- How do we determine whether a system is observable?

Finding an equation describing output derivatives

- Consider a brute-force approach. Suppose we have the model:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t)\\ z(t) &= Cx(t) + Du(t),\end{aligned}$$

and we have initial conditions $z(0)$, $\dot{z}(0)$, $\ddot{z}(0)$, $u(0)$, $\dot{u}(0)$, and $\ddot{u}(0)$.
- How can we use these to find $x(0)$?

$$\begin{aligned}z(0) &= Cx(0) + Du(0)\\ \dot{z}(0) &= C(\underbrace{Ax(0) + Bu(0)}_{\dot{x}(0)}) + D\dot{u}(0) = CAx(0) + CBu(0) + D\dot{u}(0)\\ \ddot{z}(0) &= CA^2x(0) + CABu(0) + CB\dot{u}(0) + D\ddot{u}(0).\end{aligned}$$

- In general (where superscript parentheses indicate derivatives, not powers),

$$z^{(k)}(0) = CA^kx(0) + CA^{k-1}Bu(0) + \cdots + CBu^{(k-1)}(0) + Du^{(k)}(0).$$

The observability matrix $\mathcal{O}$

- We can write this compactly in matrix form (for $n = 3$):

$$\begin{bmatrix} z(0)\\ \dot{z}(0)\\ \ddot{z}(0)\end{bmatrix} = \underbrace{\begin{bmatrix} C\\ CA\\ CA^2\end{bmatrix}}_{\mathcal{O}(C,A)} x(0) + \underbrace{\begin{bmatrix} D & 0 & 0\\ CB & D & 0\\ CAB & CB & D\end{bmatrix}}_{\mathcal{T}} \begin{bmatrix} u(0)\\ \dot{u}(0)\\ \ddot{u}(0)\end{bmatrix},$$

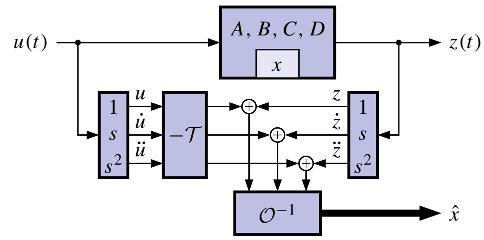
where $\mathcal{T}$ is a (block) "Toeplitz matrix" (in general, $\mathcal{O}$ has n (block) rows).
- Thus, if the observability matrix $\mathcal{O}$ is invertible, then:

$$x(0) = \mathcal{O}^{-1}\left\{\begin{bmatrix} z(0)\\ \dot{z}(0)\\ \ddot{z}(0)\end{bmatrix} - \mathcal{T}\begin{bmatrix} u(0)\\ \dot{u}(0)\\ \ddot{u}(0)\end{bmatrix}\right\}.$$

- We say that $\{C, A\}$ is an observable pair if $\mathcal{O}$ is nonsingular (for multi-input multi-output systems, $\mathcal{O}$ must be full rank).

A brute-force continuous-time observer

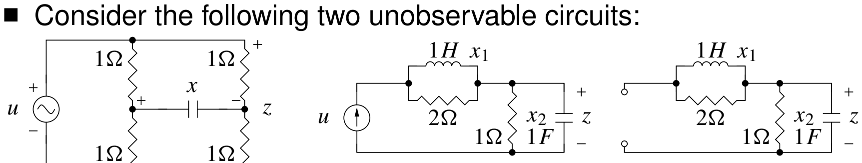
- One possible approach to determine the system state, directly from the equations, uses differentiators.
- A big problem is that differentiators amplify noise, corrupting the state estimate.
- The KF is a more practical observer that doesn't use differentiators.
- Regardless of the approach, it turns out that the system must be observable to be able to determine its initial state.



- **CONCLUSION:** If $\mathcal{O}$ is nonsingular, we can determine/estimate the initial state $x(0)$ of the system using only $u(t)$ and $z(t)$ (and so we can estimate $x(t)$ for all $t \geq 0$).
- **ADVANCED TOPIC:** If some states are unobservable but stable, an observer will still converge to the true state, although $x(0)$ may not be uniquely determined.

Examples of unobservable models

- Consider the following two unobservable circuits:



(Redrawn)

- The state-space model for the first circuit is:

$$\dot{x}(t) = -\frac{1}{C}x(t) + \frac{1}{C}u(t)$$
$$z(t) = u(t).$$

- Notice that $z(t)$ it is not a function of $x(t)$. The state-space model output equation has C matrix equal to zero. Therefore, $\mathcal{O} = 0$. Not observable.
- In the second circuit, if $u(t) = 0$, $x_1(0) \neq 0$ and $x_2(0) = 0$, then $z(t) = 0$ and we cannot determine $x_1(0)$ (circuit redrawn for $u(t) = 0$).

Controllability: Can I get there from here?

- Controllability is a *dual* idea to observability. We won't go into depth here since it is not as important for our topic of study.
- Controllability asks the question, "Can I move from any initial state to any desired state via suitable selection of the control input $u(t)$?"
- The answer boils down to a condition on a matrix called the controllability matrix

$$\mathcal{C} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}.$$

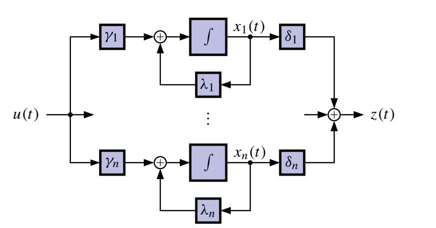
- **TEST:** If $\mathcal{C}$ is nonsingular, then the system is controllable.

Diagonal systems, controllability and observability

- We can gain insight by considering a system in diagonal form:

$$\dot{x}(t) = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} x(t) + \begin{bmatrix} \gamma_1 \\ \vdots \\ \gamma_n \end{bmatrix} u(t)$$

$$z(t) = \begin{bmatrix} \delta_1 & \cdots & \delta_n \end{bmatrix} x(t) + \begin{bmatrix} 0 \end{bmatrix} u(t).$$



- When controllable? When observable?

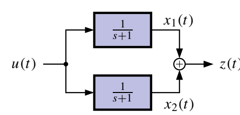
$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} = \begin{bmatrix} \delta_1 & \cdots & \delta_n \\ \lambda_1\delta_1 & \cdots & \lambda_n\delta_n \\ & \ddots & \\ \lambda_1^{n-1}\delta_1 & \cdots & \lambda_n^{n-1}\delta_n \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \cdots & 1 \\ \lambda_1 & \cdots & \lambda_n \\ & \ddots & \\ \lambda_1^{n-1} & \cdots & \lambda_n^{n-1} \end{bmatrix}}_{\text{Vandermonde matrix } \mathcal{V}} \begin{bmatrix} \delta_1 & & & 0 \\ & \delta_2 & & \\ & & \ddots & \\ 0 & & & \delta_n \end{bmatrix}.$$

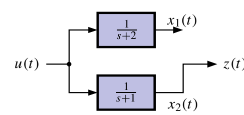
Why is a system unobservable? Uncontrollable?

- Is the observability matrix singular?

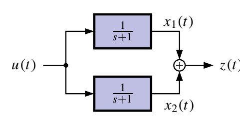
$$\det\{\mathcal{O}\} = (\delta_1 \cdots \delta_n)\det\{\mathcal{V}\} = (\delta_1 \cdots \delta_n)\prod_{i<j}(\lambda_j - \lambda_i).$$

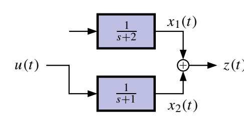
  - Observable $\Longleftrightarrow \lambda_i \neq \lambda_j,\ i \neq j$ and $\delta_i \neq 0\ i = 1, \cdots, n$.
- If $\lambda_1 = \lambda_2$ then not observable. Can only "observe" the sum $x_1 + x_2$.
- If $\delta_k = 0$ then cannot observe mode k.





- What about controllability? Analysis is similar: just switch the roles of the δs and γs.
  - Controllable $\Longleftrightarrow \lambda_i \neq \lambda_j,\ i \neq j$ and $\gamma_i \neq 0\ i = 1, \cdots, n$.
- If $\lambda_1 = \lambda_2$ then not controllable. Can only "control" the sum $x_1 + x_2$.
- If $\gamma_k = 0$ then cannot control mode k.





Equations describing discrete-time outputs

- Can we reconstruct x_0 from the output z_k and input u_k?

$$\begin{aligned} z_k &= Cx_k + Du_k \quad \text{so,} \quad z_0 = Cx_0 + Du_0 \\ z_1 &= C\left[Ax_0 + Bu_0\right] + Du_1 = CAx_0 + CBu_0 + Du_1 \\ &\vdots \\ z_{n-1} &= C\left[A^{n-1}x_0 + A^{n-2}Bu_0 + \cdots + Bu_{n-1}\right] + Du_{n-1}. \end{aligned}$$

- In matrix/vector form, we can write:

$$\begin{bmatrix} z_0 \\ z_1 \\ \vdots \\ z_{n-1} \end{bmatrix} = \underbrace{\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}}_{\mathcal{O}} x_0 + \underbrace{\begin{bmatrix} D & 0 & \cdots & 0 \\ CB & D & \cdots & 0 \\ CAB & CB & \cdots & 0 \\ \vdots & \vdots & \ddots & D \end{bmatrix}}_{\mathcal{T}} \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{n-1} \end{bmatrix}.$$

Discrete-time observability matrix $\mathcal{O}$

- So, similar to continuous-time, we write:

$$x_0 = \mathcal{O}^{-1}\left[\begin{bmatrix} z_0 \\ \vdots \\ z_{n-1}\end{bmatrix} - \mathcal{T}\begin{bmatrix} u_0 \\ \vdots \\ u_{n-1}\end{bmatrix}\right].$$

- If $\mathcal{O}$ is invertible, x_0 may be found for any z_k, u_k, and so the system is observable; also, we say that $\{C, A\}$ forms an observable pair.
- Do more measurements of $z_n, z_{n+1}, \ldots$ help in reconstructing x_0?
  - No! (Advanced topic: the Caley–Hamilton theorem).
  - So, if the original state is not observable with n measurements, then it will not be observable with more than n measurements either.
  - There is a structural problem where either a state is not connected (at all) to the output, or multiple states have the same eigenvalues (time constants).

A brute-force discrete-time observer

- Since we know u_k and the dynamics of the system, if the system is observable we can determine the entire state sequence x_k, $k \geq 0$ once we determine x_0:

$$\begin{aligned} x_n &= A^n x_0 + \sum_{i=0}^{n-1} A^{n-1-i} B u_k \\ &= A^n \mathcal{O}^{-1}\left[\begin{bmatrix} z_0 \\ \vdots \\ z_{n-1}\end{bmatrix} - \mathcal{T}\begin{bmatrix} u_0 \\ \vdots \\ u_{n-1}\end{bmatrix}\right] + \mathcal{C}\begin{bmatrix} u_{n-1} \\ \vdots \\ u_0\end{bmatrix}.\end{aligned}$$

- This is a perfectly good observer! (no differentiators...).
- But it is still not nearly as good as the Kalman filters we will develop when there is noise present.

Discrete-time controllability

- On the previous slide, I condensed the convolution sum

$$\sum_{i=0}^{n-1} A^{n-1-i} B u_k = \mathcal{C}\begin{bmatrix} u_{n-1} \\ \vdots \\ u_0\end{bmatrix},$$

  using a discrete-time controllability matrix, which I now define explicitly.
- The discrete-time controllability matrix is formed by (where we use the discrete-time A and B matrices):

$$\mathcal{C} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix}.$$

- The matrix $\mathcal{C}$ is invertible iff the system is controllable.
- Similar concept for discrete-time compared to continuous-time (and again, we won't make much use of it in this specialization).

Summary

- Is it even possible to estimate the state of our model?
- The (continuous- or discrete-time) observability matrix $\mathcal{O}$ tells us whether the initial state can be found using measurements of input and output.
  - If $\mathcal{O}$ is invertible, the system is observable and we can find $x(0)$ or x_0.
  - If $\mathcal{O}$ is not invertible, we cannot find $x(0)$ or x_0 uniquely because either a state is not connected to the output or multiple states have the same eigenvalues.
- If we can find the initial state, we can compute it at any other time. Otherwise, we cannot compute the state uniquely at any other time.
- Therefore, we will say/assume that it is necessary for our models be observable for the KF to be able to estimate the state.[1]

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[1] Advanced topic: A system is detectable if all states that cannot be observed are stable. We can apply KF to a detectable system (as the unobservable states decay toward a known trajectory in the absence of process noise) but uncertainties of unobservable states will generally become large.